

Aarhus University, Department of Mathematics
Bachelor's Degree Programme in Mathematics-Economics
Winter Exam 2022/2023
Written Exam in: 550171U026 Investment and Finance
Duration: 4 hours
Open book examination, all supplementary material allowed, no internet

Exercise 1 (weight 25%)

Assume that all the assumptions of the Markowitz framework are fulfilled. Consider an economy consisting of three assets with expected annual returns given by

$$\bar{\mathbf{r}} = \begin{pmatrix} 0.30 \\ 0.20 \\ 0.15 \end{pmatrix},$$

and covariance matrix of returns

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 \end{pmatrix} = \begin{pmatrix} 0.25 & 0.12 & 0.04 \\ 0.12 & 0.16 & 0.05 \\ 0.04 & 0.05 & 0.08 \end{pmatrix}$$

Answer the following questions:

1. Compute the portfolio weights of the Minimum Variance Portfolio (MVP). In addition, compute the expected return and the variance of the MVP.

Assume now the existence of a risk-free asset with annual return equal to $r_f = 0.03$.

2. Compute the portfolio weights of the tangency portfolio.
3. Compute the expected return and the variance of the tangency portfolio.
4. Write down the equation for the efficient frontier in the $(\sigma, \bar{r})$ -space.

Assume now that all the CAPM assumptions hold.

5. What are the weights of the market portfolio?
6. Compute the “beta” of the three assets.
7. Decompose the variance of the three assets into systematic risk and unsystematic risk.

Exercise 2 (weight 25%)

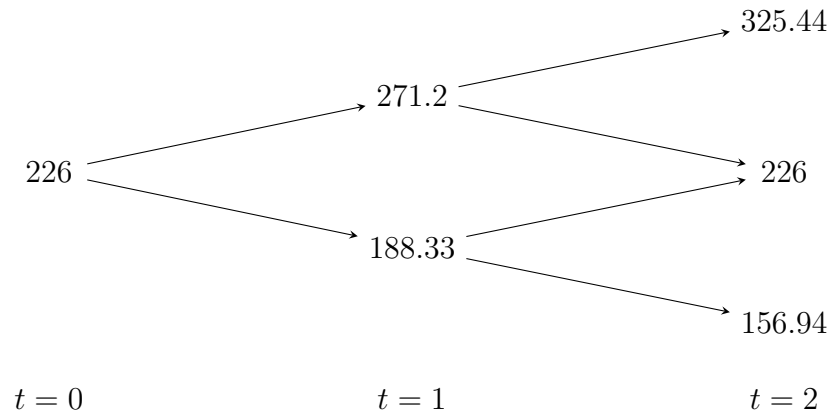
Three bonds are traded on a liquid bond market. The bonds have a face value of 100 and coupons are paid out annually. The bonds have the characteristics given in the table.

Bond	Maturity (years)	Coupon in %	Price
A	1	5.0	100.9615
B	2	3.0	98.1139
C	4	0.0	85.4804

1. Determine the spot rates s_1 , s_2 , and s_4 based on annual compounding.
2. Compute the modified duration of the three bonds.
3. Compute the convexity of the three bonds.
4. A new bond with 3 years to maturity, coupon of 4% and price 100.00 is introduced in the market. Can you use the three existing bonds to hedge a short position in the new bond against parallel changes in yields matching both present value, duration, and convexity? (Explain in detail)
5. Assume that yields shift upward by 1 percentage points. What will be the profit-loss on the hedged position derived in question 4?
6. An annuity with maturity 4 years and annual payment of 50 is introduced in the market. What is the fair price of the annuity?
7. Assume that the actual price of the annuity is 183.25. Can you propose an arbitrage strategy involving bonds A, B, C, and the bond introduced in question 4? (Explain in detail)

Exercise 3 (weight 25%)

An investment bank has calibrated the following binomial lattice describing the evolution of a non-dividend paying financial asset S over two periods:



In the financial market it is possible to trade options on S . The most liquid ones are at-the-money European call and put options with maturity at time $T = 2$. The prices of these options are, respectively

$$C_0 = 32.4198, \quad P_0 = 9.8562$$

1. What is the constant, discretely compounded interest rate r , that leads to these option prices? Assuming r is the actual interest rate, is the model then free of arbitrage?

In the following, we assume that the constant, discretely compounded interest rate r , is as found in question 1. If you did not answer question 1, make appropriate assumptions.

2. What is the American 2-year at-the-money call and put prices?
3. A new derivative contract on S enters the market. The derivative has a maturity of $T = 2$ periods and pays out $\sqrt{S(T)}$ at maturity. What is the price of this derivative today at time $t = 0$?
4. It is also possible to trade a commodity swap on S with payments falling at time $t = 1$ and $t = 2$. What is the fair price of this swap? What are the possible gains/losses with this fair price each period, assuming you pay the fixed leg?

The investment bank is interested in trading options with different maturities. Instead of calibrating the model again, they want to use a continuous-time model for S . They propose modeling S as a geometric Brownian motion as in the Black-Scholes model:

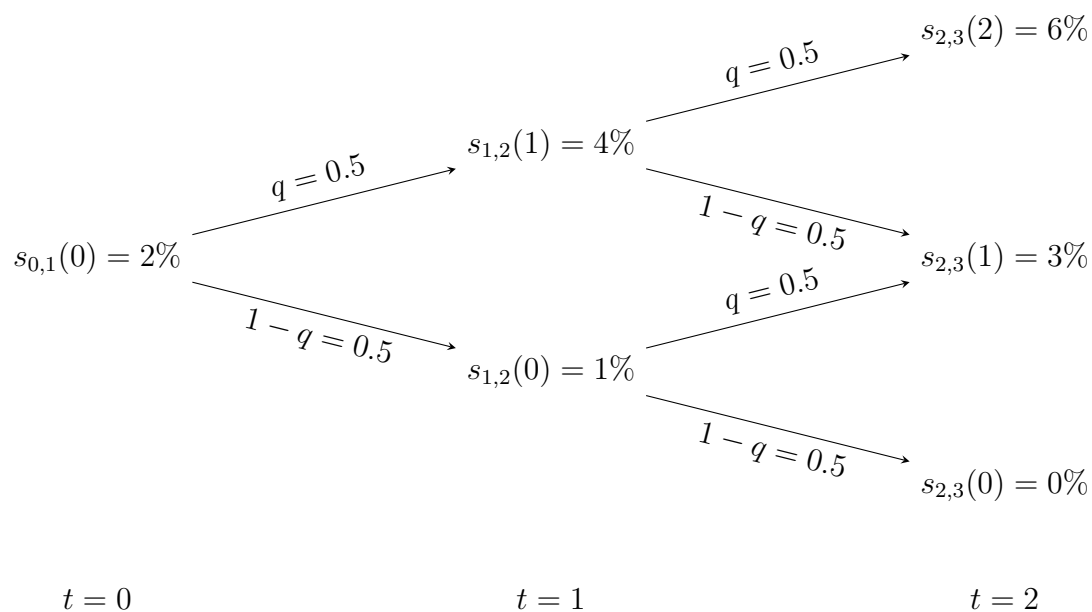
$$dS(t) = \mu S(t)dt + \sigma S(t)dW_t,$$

where W_t is a Brownian motion and $\mu, \sigma > 0$. The interest rate is still assumed to be as in question 1, although it is now assumed to be *continuously* compounded.

5. Using the approximation $u = e^{\sigma\sqrt{\Delta t}}$, what are the Black-Scholes call and put option prices for an at-the-money European option with maturity $T = 0.5$?
6. The bank owns 100 units of the asset S at time $t = 0$. They are worried about their potential losses, as they are required by authorities to have a healthy balance sheet. How can the bank use the financial market to hedge their short-term risk from owning S ?

Exercise 4 (weight 25%)

Consider the following binomial lattice model for the evolution of the short rate over 3 years:



In questions 1-5, we assume that the interest rate is discretely compounded.

1. Compute the 3-year zero coupon bond price.
2. Let $f_{s,t}$ denote the forward rate between period s and t . Compute $f_{0,1}, f_{1,2}, f_{2,3}$.
3. A 3-year coupon bond with a coupon rate of 7% is also on the market. The bond is currently trading at a price of 113.0055 per 100 notional. You are interested in buying a 2-year forward contract on this coupon bond, with takeover just after the coupon payment. How much are you willing to pay for this forward contract?
4. An investment bank offers you a 3-year interest rate swap, in which you pay the fixed leg (with yearly payments). They offer a fixed rate of $c = 0.0265$. Do you buy into this swap?
5. A friend of yours has entered into a similar 3-year interest rate swap with a notional of 100000, but with him paying the variable leg. He is afraid that interest rates will go up, so he is considering buying insurance that will pay out the difference $s(t) - K$ each period, if $s(t)$ exceeds K . He chooses to fix $K = 5\%$. How much should this insurance cost him today?
6. The same investment bank as before is also modeling interest rates in continuous time. They propose modeling the instantaneous interest rate as $r_t = x_t^2$ where

$$dx_t = te^{-\lambda t}x_t dt + \sigma e^{-\lambda t}x_t dW_t, \quad \sigma, \lambda > 0$$

What are the dynamics of r_t in this model?